Ekta Majhi

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	8.00
Kendriya Vidyalaya Shahdol (M.P.)	Class XII (CBSE)	2014	71.80%
Good Shephered Convent School, Shahdol (M.P.)	Class X (CBSE)	2012	8.60%

SCHOLASTIC ACHIEVEMENTS

Gold Medalist, National Children's Science Congress, BHU -- Recognized as a Budding Scientist for the project on biogas production.

POSITIONS OF RESPONSIBILITY

Coordinator, **DAAD Conference** -- Coordinated event activities and managed the volunteer team

SKILLS

Independent instrument operator of **Autolab** and **BioLogic** potentiostats for CV, GCD, EIS, and other electrochemical characterizations.

Independent operator of SEM (Scanning Electron Microscope), TEM (Transmission Electron Microscope), powder X-ray diffractometer (XRD), Raman spectroscopy, and Brunauer-Emmett-Teller (BET) analyzer for material characterization.

Materials Synthesis: Co-precipitation, high entropy oxide synthesis, Chemical Bath Deposition, carbon synthesis and activation, thermal/heat-treatment processes.

Electrode & Cell Fabrication: Electrode fabrication and assembly using various configurations (bath cell, swagelok cell, coin cell).

Handling instruments in a laboratory (Hot air oven, vacuum oven, freeze drier, centrifuge machine, ball milling, Hydraulic pressing machine, tubular furnace, muffle furnace, weighing balance).

RELEVANT COURSES

Material Synthesis and Characterization Techniques

Electrochemical Energy Storage Systems: Batteries, Fuel cells and Supercapacitors

Electron Microscopy

Research Ethics

Fundamentals of Electrochemistry

Practical approach to TEM

Advances in energy Storage Materials

Material Science and Engineering

Materials and Devices

Surface Science and Engineering

Computational Techniques of Materials Engineering

Technology of surface coating

2 Dimensional Materials and Electronic Devices

Electroceramics

Thin Film and Device Fabrications

PROJECTS

Co-Engineering Hierarchically Porous Carbon Electrodes and Carbon-Electrolyte Interactions to Overcome Aqueous Potential-Window Limitations for High-Energy **Supercapacitors**.

"**Gas sensors**", "Mn doped ZnO nanomaterials synthesis for gas sensing applications, response time, recovery time and stability studied

A mechanical device, **Robot arm** working on 3 DOF, controlled by "voice recognition" or "touch key pad".

EXTRACURRICULAR

National-level **Basketball** Player; Represented **IIT Hyderabad** in 3 Inter-IIT Sports Meets. Also represented school at the National-Level Basketball Championship.

SOFTWARES

Origin

NOVA 2.1

ASiQwin

EC-lab

Microsoft Office

INTERNSHIP/TRAINING/EXPERIENCE

Serving as a Teaching Assistant in the material synthesis laboratory and proficiently operating instrumentation.

Vocational training in Bharat Heavy Electricals Ltd. (**BHEL** Bhopal), manufacturing heavy mechanical components such as turbines, capacitors, heat exchangers, transformers, *etc*.

Vocational training in Hindustan Electrode Graphite Ltd. (HEG, Bhopal), manufacturing processes of graphite electrodes.

CONFERENCES

Oral presentation at the Material Research Society (MRS) conference fall meeting in 2023, in Boston, Massachusetts, USA (Utilizing Hierarchically Porous Carbon for Enhancing Supercapacitor Performance)

Poster presentation in "Gas Sensing Applications of Mn-Doped ZnO: Investigating Response Time, Recovery Time, and Stability" at the 2nd Indian Materials Conclave and 31st Annual General Meeting (AGM) in Kolkata, organized by the Material Research Society of India (MRSI) at the CSIR - Central Glass and Ceramic Research Institute, Kolkata.